STT 861: Theory of Probability & Statistics I

Lecture 18: Continuous Distributions II

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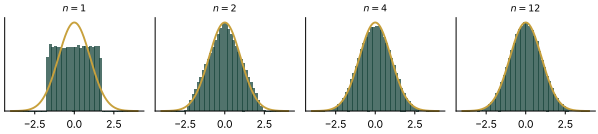
Reference: Casella & Berger Ch. 3.3

Normal: motivation — why bell curves are everywhere

- Measurement error: many small independent disturbances add up.
- Heights, test scores, noise in signals.
- Averages and sums of many things: the **Central Limit Theorem** (Ch. 5).

Preview of Chapter 5

Standardized sums of n iid Uniform$(0, 1)$ variables look more normal as n grows (right). The same happens for almost any starting distribution.



Histograms of standardized $\sum_{i=1}^{n} U_i$ (100,000 draws) with the $N(0, 1)$ density.

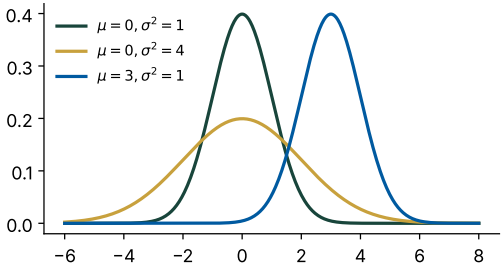
Normal(μ, σ^2): definition and parameters

Definition

$$f(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\,\sigma} \exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}, \quad x \in \mathbb{R}.$$

$\mathbb{E}X = \mu$, $\operatorname{Var} X = \sigma^2$, $M_X(t) = e^{\mu t + \sigma^2 t^2/2}$.

- Symmetric about μ; mean = median = mode.
- μ shifts the curve; σ controls spread.
- Inflection points at $\mu \pm \sigma$.



Normal: it is a pdf, and its mgf

Integrates to 1

Let $I = \int_{-\infty}^{\infty} e^{-z^2/2} dz$. Then

$$I^2 = \iint e^{-\frac{z_1^2+z_2^2}{2}} dz_1 dz_2 = \int_0^{2\pi} \int_0^{\infty} e^{-\frac{r^2}{2}} r \, dr \, d\theta = 2\pi.$$

So $\int \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz = 1$ (polar coordinates).

mgf by completing the square

For $Z \sim N(0,1)$:

$$\begin{aligned} M_Z(t) &= \int \frac{e^{tz - z^2/2}}{\sqrt{2\pi}} dz \\ &= e^{t^2/2} \int \frac{e^{-(z-t)^2/2}}{\sqrt{2\pi}} dz = e^{t^2/2}. \end{aligned}$$

For $X = \mu + \sigma Z$: $M_X(t) = e^{\mu t} M_Z(\sigma t) = e^{\mu t + \sigma^2 t^2/2}$.

Moments from the mgf

$M_Z'(0) = 0$, $M_Z''(t) = (1 + t^2) e^{t^2/2} \Rightarrow \mathbb{E}Z^2 = 1$. So $\mathbb{E}Z = 0$, $\mathrm{Var}\, Z = 1$, and $\mathbb{E}X = \mu$, $\mathrm{Var}\, X = \sigma^2$.

Standardization and the 68–95–99.7 rule

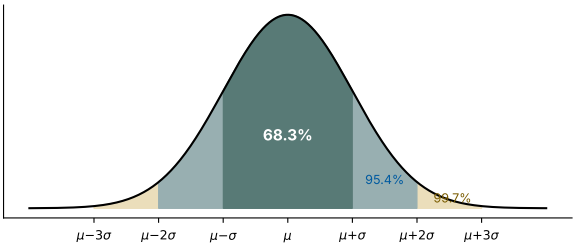
Standardization

$X \sim N(\mu, \sigma^2) \iff Z = \dfrac{X-\mu}{\sigma} \sim N(0,1)$,
and so

$$P(X \le x) = \Phi\Big(\frac{x-\mu}{\sigma}\Big).$$

Linear transformations

$Y = aX + b \sim N(a\mu + b,\, a^2\sigma^2)$ since $M_Y(t) = e^{bt} M_X(at) = e^{(a\mu+b)t + a^2\sigma^2 t^2/2}$.



$P(\lvert Z\rvert \le 1)$	0.683	$\Phi(1.645)$	0.950
$P(\lvert Z\rvert \le 2)$	0.954	$\Phi(1.96)$	0.975
$P(\lvert Z\rvert \le 3)$	0.997	$\Phi(2.326)$	0.990

Example: IQ scores

Scores $X \sim N(100, 15^2)$

(a) $P(X > 130) = P(Z > 2) = 1 - \Phi(2) \approx 0.0228$.

(b) $P(85 < X < 115) = P(-1 < Z < 1) \approx 0.683$.

(c) The 90th percentile: $\Phi(z) = 0.90 \Rightarrow z \approx 1.2816$, so $x = 100 + 15(1.2816) \approx 119.2$.

(d) A test taker is converted to $Y = \dfrac{X - 100}{15} \cdot 10 + 50$. Distribution of Y?
By the linear rule $Y \sim N\big(50,\, 10^2\big)$ (this is the usual "T-score").

Pair discussion

If $X \sim N(\mu, \sigma^2)$, is $P(|X - \mu| < 2\sigma)$ the same for every μ and σ? *(Yes, ≈ 0.954: standardization removes both.)*

Activity 1— Normal calculations

A filling machine: bottle contents $X \sim N(16.0,\ 0.2^2)$ oz

(a) Find $P(X < 15.7)$ (use $\Phi(1.5) \approx 0.9332$).

(b) Find c such that $P(|X - 16| < c) = 0.95$ (use $z_{0.975} = 1.96$).

(c) A bottle is relabeled as $Y = 3 - 2X$. Give the distribution of Y, $\mathbb{E}Y$ and $\mathrm{Var}\, Y$.

(d) Two bottles X_1, X_2 are filled independently. Using mgfs, find the distribution of $X_1 + X_2$.

Activity 2— Solutions

(a) $P(X < 15.7) = \Phi\left(\frac{15.7-16}{0.2}\right) = \Phi(-1.5) = 1 - 0.9332 = 0.0668.$

(b) $P(|Z| < c/0.2) = 0.95 \Rightarrow c/0.2 = 1.96 \Rightarrow c = 0.392$ oz.

(c) $Y = 3 - 2X \sim N\big(3 - 2(16),\ (-2)^2(0.04)\big) = N(-29,\ 0.16)$: $\mathbb{E}Y = -29$, $\text{Var}\, Y = 0.16$.

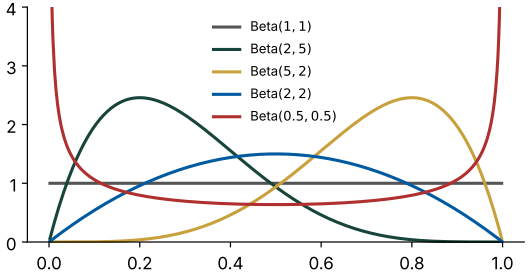
(d) $M_{X_1+X_2}(t) = M_{X_1}(t)M_{X_2}(t) = e^{32t+0.04t^2}$, i.e. $N(32,\ 0.08)$: means add, variances add.

Beta: motivation — modelling a proportion

- Fraction of defective parts in a batch; click-through rate; share of votes.
- All are numbers in $(0, 1)$ and we want flexible shapes: U-shaped, flat, skewed, bell-like.
- In Bayesian statistics, a **prior** for an unknown success probability p (preview of §7.2).

Beta function

$$B(\alpha, \beta) = \int_0^1 x^{\alpha-1}(1-x)^{\beta-1}dx = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}.$$



Beta(α, β): definition and moments

Definition ($\alpha, \beta > 0$)

$$f(x \mid \alpha, \beta) = \frac{1}{B(\alpha, \beta)} x^{\alpha-1}(1-x)^{\beta-1}, \quad 0 < x < 1.$$

Beta$(1, 1)$ is Uniform$(0, 1)$.

Moments

$$\mathbb{E}X^n = \frac{B(\alpha+n, \beta)}{B(\alpha, \beta)} = \frac{\Gamma(\alpha+n)\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\alpha+\beta+n)}.$$

$$\mathbb{E}X = \frac{\alpha}{\alpha+\beta}, \qquad \text{Var}\, X = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}.$$

Derivation of the mean

$\mathbb{E}X = \frac{B(\alpha+1,\beta)}{B(\alpha,\beta)} = \frac{\Gamma(\alpha+1)\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\alpha+\beta+1)} = \frac{\alpha}{\alpha+\beta}$, using $\Gamma(a+1) = a\Gamma(a)$ twice. Same idea: *the kernel integrates to a normalizing constant.*

Beta: examples

Example 1: $X \sim \mathrm{Beta}(2,5)$

$B(2,5) = \frac{\Gamma(2)\Gamma(5)}{\Gamma(7)} = \frac{1\cdot 24}{720} = \frac{1}{30}$, so $f(x) = 30\,x(1-x)^4$.

- $\mathbb{E}X = \frac{2}{7} \approx 0.286$, $\mathbb{E}X^2 = \frac{2\cdot 3}{7\cdot 8} = \frac{3}{28}$.
- $\mathrm{Var}\,X = \frac{3}{28} - \frac{4}{49} = \frac{10}{392} \approx 0.0255$.
- $P(X < \frac{1}{2}) = \int_0^{1/2} 30x(1-x)^4\,dx = \frac{57}{64} \approx 0.891$.

Example 2 (Bayes preview)

Prior $p \sim \mathrm{Beta}(a,b)$; data $x \mid p \sim \mathrm{Bin}(n,p)$. Multiply the kernels:

$$\pi(p \mid x) \propto p^{a+x-1}(1-p)^{b+n-x-1}.$$

So $p \mid x \sim \mathrm{Beta}(a+x,\ b+n-x)$.

With $a = b = 2$, $n = 10$, $x = 7$: posterior $\mathrm{Beta}(9,5)$, mean $\frac{9}{14} \approx 0.643$ (prior mean 0.5).

Takeaway

The Beta is the *conjugate* family for the Binomial.

Cauchy: motivation — heavy tails

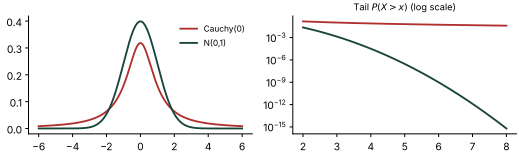
Cauchy(θ)

$$f(x \mid \theta) = \frac{1}{\pi\,[1 + (x - \theta)^2]}, \quad x \in \mathbb{R}.$$

Symmetric about θ (the median), looks like a normal but with much fatter tails.

Where it arises

- Z_1/Z_2 for independent $N(0,1)$ variables is Cauchy(0).
- Flashlight at angle $X \sim U(-\frac{\pi}{2}, \frac{\pi}{2})$ hitting a wall: $Y = d \tan X$ has pdf $\frac{d}{\pi(d^2+y^2)}$ (Exam I).



Cauchy has no mean — and no mgf

$\mathbb{E}|X| = \infty$

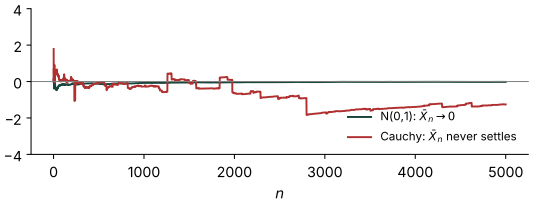
For $\theta = 0$:

$$\mathbb{E}|X| = 2\int_0^\infty \frac{x}{\pi(1+x^2)}\,dx = \frac{1}{\pi}\left[\log(1+x^2)\right]_0^\infty = \infty.$$

Hence $\mathbb{E}X$ does **not exist**, $\mathrm{Var}\,X$ does not exist, and no mgf exists on any neighbourhood of 0.

Careful

By symmetry one might guess "$\mathbb{E}X = 0$". The integral $\int_{-\infty}^{\infty} x f(x)dx$ is not defined: $\infty - \infty$.



Running sample mean: the Cauchy one jumps whenever an extreme value arrives. The sample mean of n Cauchy(θ) draws is again Cauchy(θ), so averaging does not help (the law of large numbers fails without a mean).

Lognormal: multiplicative growth

Definition

$Y = e^X$ where $X \sim N(\mu, \sigma^2)$ (equivalently $\log Y \sim N(\mu, \sigma^2)$). For $y > 0$:

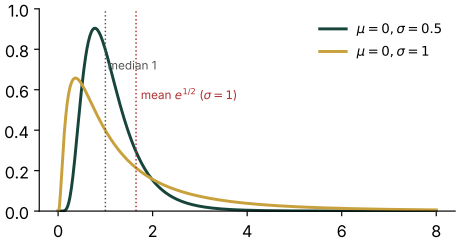
$$f_Y(y) = \frac{1}{\sqrt{2\pi}\,\sigma\, y} \exp\!\left\{-\frac{(\log y - \mu)^2}{2\sigma^2}\right\}.$$

Moments from the normal mgf

$\mathbb{E}Y = \mathbb{E}e^X = M_X(1) = e^{\mu+\sigma^2/2}$, $\mathbb{E}Y^2 = M_X(2) = e^{2\mu+2\sigma^2}$,

$$\operatorname{Var} Y = e^{2(\mu+\sigma^2)} - e^{2\mu+\sigma^2}.$$

Median $= e^{\mu} <$ mean: right-skewed.



Why use it?

Stock prices, incomes, lifetimes: growth by *multiplying* many random factors ⇒ the log is a sum ⇒ approximately normal.

Example: lognormal, and double exponential

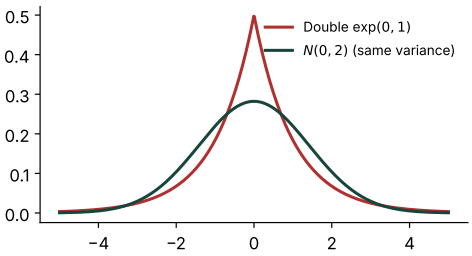
Lognormal with $\mu = 0, \sigma = 1$

- $\mathbb{E}Y = e^{1/2} \approx 1.649$, median $= 1$.
- $\operatorname{Var} Y = e^2 - e \approx 4.671$.
- $P(Y > e) = P(X > 1) = 1 - \Phi(1) \approx 0.159$.

Double exponential(μ, σ)

$$f(x) = \frac{1}{2\sigma} e^{-|x-\mu|/\sigma}, \quad x \in \mathbb{R}.$$

$\mathbb{E}X = \mu$, $\operatorname{Var} X = 2\sigma^2$, $M_X(t) = \dfrac{e^{\mu t}}{1 - \sigma^2 t^2}$, $|t| < 1/\sigma$.



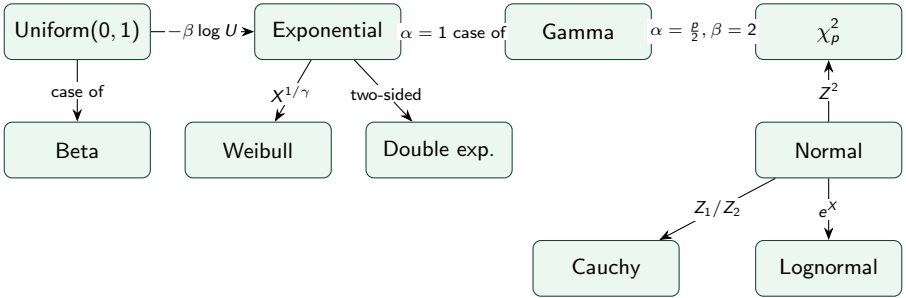
Connections

$|X - \mu| \sim \text{Exponential}(\sigma)$, so the double exponential is a "two-sided exponential". Its sharp peak and heavier tails (than normal) make it a model for outlier-prone noise (and the basis of L_1 / median regression and the lasso penalty).

Summary of §3.3: the catalogue

Family	**pdf kernel**	**Mean**	**Variance**	**mgf**
Uniform(a, b)	1 on $[a, b]$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$	$\frac{e^{bt}-e^{at}}{(b-a)t}$
Gamma(α, β)	$x^{\alpha-1}e^{-x/\beta}$	$\alpha\beta$	$\alpha\beta^2$	$(1-\beta t)^{-\alpha}$
Exponential(β)	$e^{-x/\beta}$	β	β^2	$(1-\beta t)^{-1}$
χ^2_p	$x^{p/2-1}e^{-x/2}$	p	$2p$	$(1-2t)^{-p/2}$
Normal(μ, σ^2)	$e^{-(x-\mu)^2/2\sigma^2}$	μ	σ^2	$e^{\mu t+\sigma^2 t^2/2}$
Beta(α, β)	$x^{\alpha-1}(1-x)^{\beta-1}$	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$	(not closed form)
Cauchy(θ)	$[1+(x-\theta)^2]^{-1}$	does not exist	does not exist	does not exist
Lognormal(μ, σ^2)	$\frac{1}{y}e^{-(\log y-\mu)^2/2\sigma^2}$	$e^{\mu+\sigma^2/2}$	$e^{2(\mu+\sigma^2)}-e^{2\mu+\sigma^2}$	(does not exist)
Double exp.(μ, σ)	$e^{-\lvert x-\mu\rvert/\sigma}$	μ	$2\sigma^2$	$\frac{e^{\mu t}}{1-\sigma^2 t^2}$

How the families connect



The "$\rightarrow$" means *special case* or *transformation of*.

Activity 2 — Beta, lognormal, Cauchy

Short problems

(a) $X \sim \mathrm{Beta}(3, 2)$. Write its pdf explicitly (compute $B(3, 2)$), find $\mathbb{E}X$, $\mathrm{Var}\, X$ and $P(X > \frac{1}{2})$.

(b) $\log Y \sim N(1, 4)$. Find the median of Y and $\mathbb{E}Y$.

(c) $X_1, \ldots, X_{100}$ iid Cauchy(0). True or false, and why: "$\bar{X}$ has a smaller spread than a single X_1."

(d) A company's weekly sales are the *product* of 52 independent positive growth factors. Which family would you try for the yearly total factor, and why?

Activity 2— Solutions

(a) $B(3,2) = \frac{\Gamma(3)\Gamma(2)}{\Gamma(5)} = \frac{2}{24} = \frac{1}{12}$, so $f(x) = 12x^2(1-x)$. $\mathbb{E}X = \frac{3}{5}$, $\operatorname{Var} X = \frac{6}{25 \cdot 6} = \frac{1}{25} = 0.04$.
$P(X > \frac{1}{2}) = 12\left[\frac{x^3}{3} - \frac{x^4}{4}\right]_{1/2}^{1} = 1 - \frac{5}{16} = \frac{11}{16}$.

(b) Median $= e^1 \approx 2.718$. $\mathbb{E}Y = e^{\mu + \sigma^2/2} = e^{1+2} = e^3 \approx 20.09$ (mean $\gg$ median: right skew).

(c) **False.** $\bar{X}$ is again Cauchy(0): same spread as one observation. No mean means no LLN.

(d) Lognormal: log of the product is a sum of 52 independent terms, approximately normal by the CLT, so the product is approximately lognormal.

Activity 3— Match the story to a family

1. Time until the next earthquake in a region (memoryless).
2. Annual household incomes in a city.
3. The proportion of a population that supports a measure.
4. Total time until the 10th customer arrives.
5. Sum of squares of 7 independent standardized measurement errors.
6. Ratio of two independent, centered measurement errors.
7. A random number from a generator on $(0, 1)$.
8. Lifetime of a component that wears out over time.

Choices

A. Uniform B. Exponential C. Gamma
D. χ^2 E. Normal F. Beta G. Cauchy
H. Lognormal I. Weibull

Activity 3— Solutions

1. B (Exponential) 2. H (Lognormal) 3. F (Beta) 4. C (Gamma, $\alpha = 10$)
2. D (χ^2_7) 6. G (Cauchy) 7. A (Uniform$(0, 1)$) 8. I (Weibull, $\gamma > 1$)

Wrap-up

- Continuous families are tied together by transformations and by the mgf.
- Always ask: *does the mean exist?* (Cauchy: no.)
- Next: §3.4 exponential families, and then Chapter 4 (joint distributions).

Homework: CB Ch. 3 exercises on the Gamma, Normal and Beta families.